

Huffman code: "ROSES ARE RED AND VIOLETS ARE BLUE"

(spaces ignored $\Rightarrow$ 28 letters)

a) Letter counts: Sum=28

c) Codes:

E: 6, R: 4, A: 3

S: 3, D: 2, L: 2

O: 2, B: 1, I: 1

N: 1, T: 1, U: 1, V: 1

E: 00 $\Rightarrow 6 \times 2 = 12$,

R: 100 $\Rightarrow 4 \times 3 = 12$,

A: 010 $\Rightarrow 3 \times 3 = 9$,

S: 011 $\Rightarrow 3 \times 3 = 9$,

D: 1010 $\Rightarrow 2 \times 4 = 8$,

L: 1011 $\Rightarrow 2 \times 4 = 8$, O: 1100 $\Rightarrow 2 \times 4 = 8$, B: 11010 $\Rightarrow 1 \times 5 = 5$,

I: 11011 $\Rightarrow 1 \times 5 = 5$, N: 11100 $\Rightarrow 1 \times 5 = 5$, T: 11101 $\Rightarrow 1 \times 5 = 5$,

U: 11110 $\Rightarrow 1 \times 5 = 5$, V: 11111 $\Rightarrow 1 \times 5 = 5$,

b) Tree (left=0, Right=1):

Merge order:

B+I=2, N+T=2,

U+V=2,

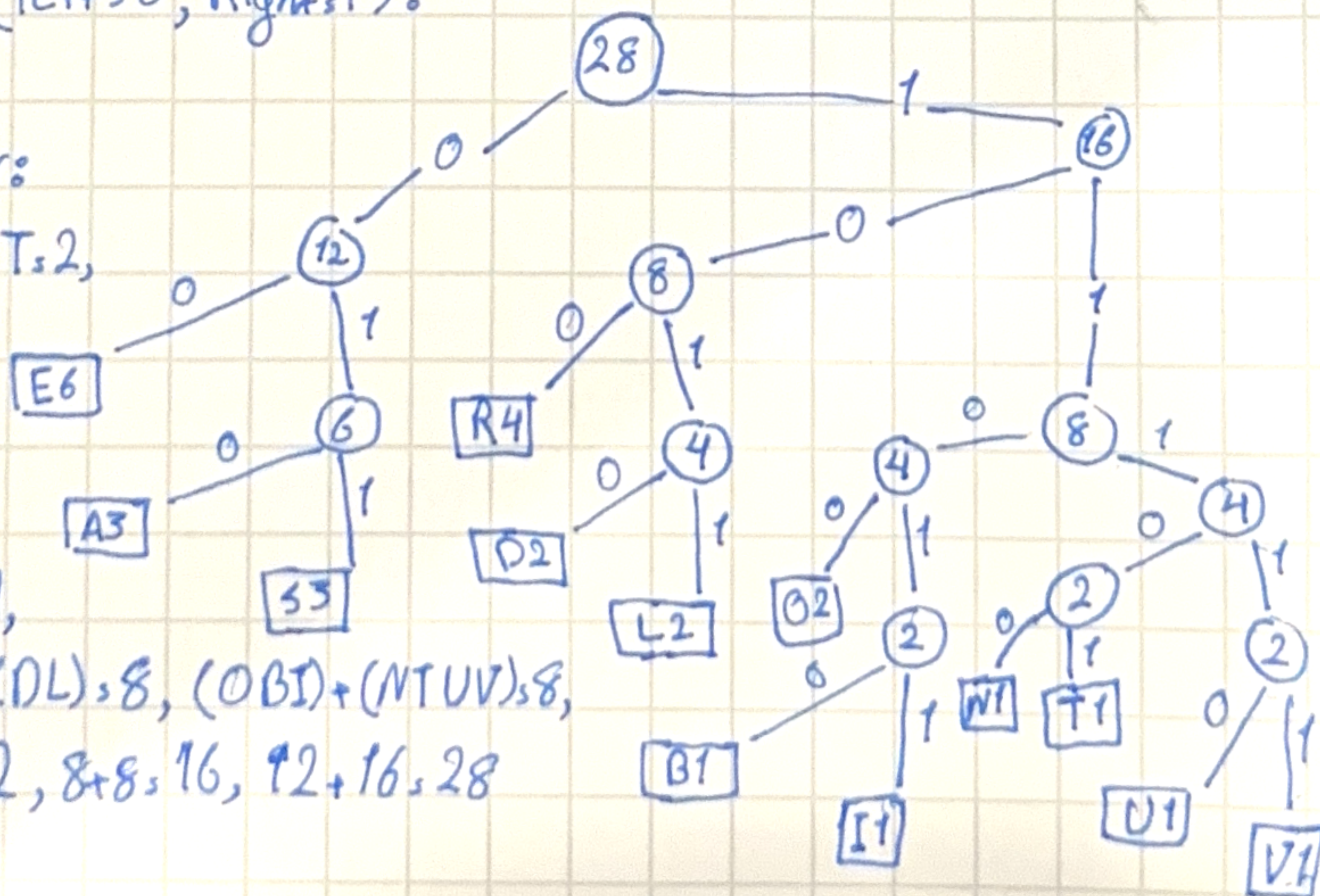
D+L=4,

O+(BI)=4,

(NT)+(UV)=4,

A+S=6, R+(DL)=8, (OBI)+(NTUV)=8,

E+(AS)=12, 8+8=16, 12+16=28



d) 1. 13 letters

2. 4 bits ($2^4 = 16 > 13$)

3. $28 \times 4 = 112$ bits

4. 96 bits (-14%)